Practice Questions for 2024-10-28-ApplsLinProg

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1 Questions

1. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Why is this term included, and what is its practical significance in the context of ice cream production?

   a. The term is included to penalize frequent changes in production, which can be costly due to setup times and resource reallocation. This ensures a more stable and efficient production plan.

   b. The term is included to model the seasonal fluctuations in ice cream demand, ensuring that production closely matches the demand in each month.

   c. The term is included to simplify the linear programming model, making it easier to solve. It does not have any practical significance in the context of ice cream production.

   d. The term is included to account for potential spoilage of ice cream if production exceeds demand. This ensures that surplus inventory is minimized.

   e. The term is included to model the cost of storing surplus ice cream, ensuring that the total storage cost is minimized.

2. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and why this reformulation is necessary.

   a. The problem is reformulated using a logarithmic transformation to convert the max function into a linear expression. This is necessary to apply standard linear programming techniques.

   b. The problem is reformulated by introducing auxiliary variables and constraints to represent the max function linearly. This is necessary because standard linear programming algorithms cannot handle non-linear objective functions.

   c. The problem is reformulated by approximating the max function using a piecewise linear function. This is necessary to improve the accuracy of the solution.

   d. The problem is reformulated by ignoring the max function, simplifying the objective function to a linear expression. This is necessary to reduce the computational complexity of the problem.

   e. The problem is reformulated using a quadratic approximation of the max function. This is necessary to handle the non-linearity of the max function while maintaining computational efficiency.

3. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

a. The model is transformed into a linear program by ignoring the randomness of the returns and using the actual returns from the previous year as fixed values.

b. The model is transformed into a quadratic program by replacing the random returns with their expected values and using the covariance matrix to represent the risk.

c. The model is transformed into a linear program by approximating the returns using a normal distribution and using the mean and standard deviation as fixed values.

d. The model is transformed into a stochastic program by explicitly incorporating the probability distributions of the returns into the optimization problem.

e. The model is transformed into a deterministic program by assuming that all returns are equal to their median values.

4. The Cutting Stock Problem (Slides 25-30) involves finding the optimal way to cut large rolls of paper into smaller rolls to meet customer orders while minimizing waste. Explain why this problem is typically formulated as an integer programming problem, and what challenges this introduces compared to standard linear programming.

   a. The problem is formulated as an integer program because the number of cuts must be an integer. This introduces computational challenges because integer programming problems are generally more difficult to solve than linear programs.

   b. The problem is formulated as an integer program because the widths of the smaller rolls are integers. This introduces challenges because the feasible region becomes discrete, making it harder to find the optimal solution.

   c. The problem is formulated as an integer program because the number of smaller rolls of each width must be an integer. This introduces challenges because the feasible region becomes non-convex, making it harder to find the optimal solution.

   d. The problem is formulated as an integer program because the total length of the large roll is an integer. This introduces challenges because the constraints become non-linear, making it harder to find the optimal solution.

   e. The problem is formulated as an integer program because the number of large rolls used must be an integer. This introduces challenges because the objective function becomes non-linear, making it harder to find the optimal solution.

5. a. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and discuss its practical significance in the context of ice cream production.

      A. This term accounts for the fixed costs associated with starting and stopping production lines, which are independent of the production volume.

      B. This term reflects the cost of labor and materials, which are directly proportional to the amount of ice cream produced each month.

      C. This term represents the cost of storing surplus ice cream from one month to the next, which increases with the amount of stored ice cream.

      D. This term models the fluctuating demand for ice cream throughout the year, which necessitates adjustments in production levels.

      E. This term captures the inherent variability in the quality of ice cream produced, which is affected by changes in production rates.

b. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and justify why this reformulation is necessary.

   A. The max function is replaced by a set of binary variables, representing whether or not to use temporary employees in each month.

   B. The max function is approximated using a piecewise linear function, dividing the range of possible values into intervals.

   C. The max function is replaced by an auxiliary variable and a set of linear constraints that ensure the auxiliary variable is greater than or equal to the arguments of the max function.

   D. The max function is eliminated by considering only the worst-case scenario, where the maximum demand is always met using temporary employees.

   E. The max function is replaced by its expected value, assuming a probability distribution for the demand.

c. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and identify the key assumptions or approximations made in this transformation.

   A. The random returns are replaced by their worst-case values, leading to a robust optimization problem.

   B. The random returns are replaced by their most likely values, based on historical data.

   C. The random returns are replaced by their expected values, and the variance of the returns is incorporated into the objective function as a measure of risk.

   D. The random returns are modeled using a stochastic programming approach, explicitly considering the probability distribution of the returns.

   E. The random returns are ignored, and the optimization problem focuses solely on maximizing the expected return.

6. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and discuss its practical significance in the context of ice cream production.

   a. This term is included to simplify the model, making it easier to solve using linear programming techniques. It has no significant practical impact on ice cream production.

   b. This term reflects the real-world costs associated with adjusting production lines, hiring and firing workers, and managing inventory fluctuations. In ice cream production, frequent changes are expensive and inefficient.

   c. This term is a placeholder and doesn't represent any real-world cost. It's included to make the model more complex and challenging for students.

   d. This term is included to account for the seasonal nature of ice cream demand. It has no direct relation to production costs.

   e. This term is a mathematical artifact with no practical interpretation in the context of ice cream production.

7. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and justify why this reformulation is necessary.

a. The max function is replaced with an average function, approximating the original problem. This is sufficient because the error introduced is negligible.

b. The problem is reformulated using a piecewise linear approximation of the max function. This is computationally expensive but provides an exact solution.

c. The max function is replaced with an auxiliary variable and additional constraints, linearizing the problem. This is necessary because standard linear programming algorithms cannot handle non-linear objective functions.

d. The max function is ignored, simplifying the problem to a linear one. This is acceptable because the max function only slightly affects the overall cost.

e. The problem is reformulated using dynamic programming, which is more efficient than linear programming for this type of problem.

8. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

   a. The random returns are replaced by their worst-case values, resulting in a robust optimization problem. This approach is conservative but guarantees a feasible solution.

   b. The random returns are ignored, simplifying the problem to a deterministic one. This is acceptable if the variability of returns is small.

   c. The random returns are replaced by their expected values, transforming the problem into a deterministic optimization problem. This assumes that the investor is risk-neutral.

   d. The random returns are modeled using a Monte Carlo simulation, generating a large number of scenarios. The solution is then the average of the optimal solutions for each scenario.

   e. The random returns are replaced by their median values, providing a more robust solution than using the expected values. This approach is less sensitive to outliers.

9. The Cutting Stock Problem (Slides 25-30) involves finding the optimal way to cut large rolls of paper into smaller rolls to meet customer orders while minimizing waste. Explain why this problem is typically formulated as an integer programming problem, and what challenges this introduces compared to standard linear programming.

   a. The problem is formulated as an integer program because the number of rolls must be a whole number. This introduces computational challenges because integer programming problems are generally much harder to solve than linear programming problems.

   b. The problem is formulated as an integer program to simplify the model. This makes the problem easier to solve, but it may not find the optimal solution.

   c. The problem is formulated as an integer program because the cutting patterns must be chosen from a discrete set. This introduces no significant computational challenges.

   d. The problem is formulated as an integer program because the waste must be minimized. This introduces no computational challenges compared to standard linear programming.

   e. The problem is formulated as an integer program because the customer orders must be met exactly. This introduces no computational challenges compared to standard linear programming.

10. a. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and what its practical significance is in the context of ice cream production.

    A. This term is included to penalize frequent changes in production, reflecting the real-world costs associated with adjusting equipment, staffing, and inventory levels. In ice cream production, sudden shifts in production can lead to inefficiencies and increased costs.

    B. This term is a mathematical artifact with no practical significance; it simplifies the model without affecting the optimal solution.

    C. This term is included to ensure that production levels remain constant throughout the year, regardless of demand fluctuations.

    D. This term is included to model the spoilage of ice cream due to overproduction.

    E. This term is irrelevant to the optimization problem and can be safely ignored.

    b. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and why this reformulation is necessary.

    A. The problem is reformulated by introducing auxiliary variables that represent the maximum function, converting the non-linear constraints into linear ones. This is necessary because standard linear programming algorithms cannot handle non-linear objective functions.

    B. The problem is reformulated using a logarithmic transformation, which converts the max function into a linear expression. This is necessary to ensure the feasibility of the solution.

    C. The problem is reformulated by ignoring the max function, simplifying the objective function. This is necessary to reduce the computational complexity.

    D. The problem is reformulated by approximating the max function with a piecewise linear function. This is necessary to improve the accuracy of the solution.

    E. No reformulation is necessary; the problem can be solved directly using non-linear programming techniques.

    c. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

    A. The model is transformed by replacing the random returns with their expected values, resulting in a linear programming problem. The key assumption is that the returns are normally distributed.

    B. The model is transformed by using the variance of the returns as a measure of risk, resulting in a quadratic programming problem. The key assumption is that the returns are independent and identically distributed.

    C. The model is transformed by ignoring the randomness of the returns, resulting in a deterministic optimization problem. The key assumption is that the returns are constant over time.

    D. The model is transformed by using a Monte Carlo simulation to generate multiple scenarios of returns, resulting in a stochastic programming problem. The key assumption is that the returns follow a specific probability distribution.

    E. The model is transformed by using a linear approximation of the returns, resulting in a linear programming problem. The key assumption is that the returns are linearly related to the investment proportions.

2 Answers

1. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Why is this term included, and what is its practical significance in the context of ice cream production?

   a. The cost of changing production levels is included to reflect the real-world costs associated with adjusting production lines, re-training staff, and managing resource allocation. Frequent changes are inefficient and expensive. A stable production plan minimizes these costs, leading to greater overall efficiency.

   b. While the production plan should consider seasonal demand, the cost of changing production levels is not directly related to modeling the demand fluctuations. The model's constraints handle demand fulfillment.

   c. The term is not included to simplify the model; rather, it adds complexity. It accurately reflects a significant cost component in ice cream production.

   d. Spoilage is a concern, but the cost of changing production levels is not directly related to spoilage. The model addresses surplus inventory through a separate storage cost term.

   e. The cost of storing surplus ice cream is a separate term in the objective function. The cost of changing production levels is distinct and reflects the cost of adjusting production quantities.

2. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and why this reformulation is necessary.

   a. Logarithmic transformations are not typically used to linearize max functions in this context. Auxiliary variables and constraints provide a more direct and effective approach.

   b. The reformulation introduces auxiliary variables (y_i) to represent the output of the max function. Constraints are then added to ensure that these auxiliary variables correctly capture the maximum value. This is necessary because standard linear programming algorithms require a linear objective function and linear constraints.

   c. Piecewise linear approximations can be used for non-linear functions, but introducing auxiliary variables is a more precise and efficient method for linearizing the max function in this specific problem.

   d. Ignoring the max function would lead to an incorrect and potentially suboptimal solution. The max function is crucial for accurately modeling the cost of temporary employees.

   e. Quadratic approximations are not necessary or efficient in this case. The linearization using auxiliary variables is a simpler and more accurate approach.

3. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

   a. Ignoring the randomness entirely would lead to a highly inaccurate and potentially misleading solution. The model needs to account for the uncertainty in returns.

   b. The transformation replaces the random returns with their expected values, resulting in an objective function that maximizes the expected portfolio return. The risk is incorporated using the covariance matrix of the returns, leading to a quadratic program (a generalization of linear programming).

c. Approximating returns with a normal distribution is a possibility, but the presented transformation uses expected values and the covariance matrix directly.

d. While stochastic programming techniques exist, the presented transformation leads to a deterministic quadratic program by focusing on expected values and covariance.

e. Using median values instead of expected values would be an alternative approach, but the presented transformation uses expected values.

4. The Cutting Stock Problem (Slides 25-30) involves finding the optimal way to cut large rolls of paper into smaller rolls to meet customer orders while minimizing waste. Explain why this problem is typically formulated as an integer programming problem, and what challenges this introduces compared to standard linear programming.

   a. The decision variables represent the number of times each cutting pattern is used. These must be integers because you cannot use a fractional number of cutting patterns. Integer programming problems are generally NP-hard, meaning they are computationally more complex than linear programs, which can be solved in polynomial time.

   b. While the widths are integers, the key reason for using integer programming is the integer number of times each cutting pattern is used, not the integer nature of the widths themselves.

   c. The feasible region is not necessarily non-convex. The constraints are linear, but the integer requirement makes the problem more difficult to solve.

   d. The total length of the large roll is a constant and does not directly affect the integer nature of the decision variables.

   e. The total number of large rolls used is the objective function, not a constraint. The integer requirement comes from the number of times each cutting pattern is used.

5. a. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and discuss its practical significance in the context of ice cream production.

   A. The cost of changing production levels is included to reflect the real-world expenses associated with adjusting production capacity. In ice cream production, this might involve bringing additional equipment online, hiring temporary workers, or incurring other setup costs when increasing production, and conversely, laying off temporary workers or incurring costs associated with idling equipment when decreasing production. These costs are not directly proportional to the production volume but rather to the magnitude of the change in production between consecutive months. Therefore, minimizing this term in the objective function leads to a more stable production plan, reducing unnecessary fluctuations and associated costs.

   B. The cost of labor and materials is indeed a significant factor in ice cream production, but it's typically modeled as a cost directly proportional to the production volume in each month, not as a cost associated with changes in production levels between months.

   C. The cost of storing surplus ice cream is a separate term in the objective function, explicitly representing the cost of storage. The cost of changing production levels is distinct and reflects the costs associated with adjusting production capacity.

   D. Fluctuating demand is accounted for by the constraints in the model, which ensure that production plus beginning inventory meets the demand in each month. The cost of changing production levels is an additional cost incurred due to the adjustments needed to meet this fluctuating demand.

E. Variability in ice cream quality is not directly addressed by this term in the objective function. Quality control measures are likely handled separately in the production process.

b. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and justify why this reformulation is necessary.

A. While binary variables are often used in integer programming, they are not the primary method used here to linearize the max function. The auxiliary variable approach is more efficient and directly addresses the non-linearity.

B. Piecewise linear approximations are a valid approach for some non-linear functions, but they are not used in this specific reformulation. The auxiliary variable method provides an exact linearization.

C. The reformulation uses an auxiliary variable (y_i) to represent $\max(a_i - x, 0)$, where a_i is the labor hours needed in month i and x is the number of regular employee hours. The constraint $a_i - x \leq y_i$ ensures that y_i is at least as large as the expression inside the max function. Since y_i must be non-negative, the constraint effectively captures the max function. This reformulation is necessary because standard linear programming algorithms require a linear objective function and linear constraints. The original objective function, containing the non-linear max function, is not directly solvable using these algorithms. The linearization allows the application of efficient linear programming techniques to find the optimal solution.

D. Considering only the worst-case scenario would lead to an overly conservative and potentially suboptimal solution. The linearization allows for a more nuanced and optimal solution.

E. Replacing the max function with its expected value would require knowledge of the probability distribution of the demand, which is not assumed in the problem statement. The auxiliary variable method is more general and does not require such assumptions.

c. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and identify the key assumptions or approximations made in this transformation.

A. Using worst-case values would lead to a very conservative portfolio, potentially missing out on higher returns. The Markowitz model aims for a balance between risk and return.

B. While historical data is used to estimate the parameters, simply using the most likely value ignores the uncertainty and risk associated with the investments.

C. The transformation involves replacing the random returns (R_j) with their expected values (ER_j) and incorporating the variance (or covariance) of the returns into the objective function. The objective function becomes a trade-off between maximizing the expected return and minimizing the risk (variance). This transforms the problem into a quadratic programming problem (a generalization of linear programming). The key assumption is that the expected values and covariances of the returns are known or can be reliably estimated from historical data. This approach simplifies the problem by focusing on the mean and variance of the returns, ignoring higher-order moments of the return distribution. This is a common approximation in portfolio optimization.

D. Stochastic programming explicitly incorporates the probability distribution of the returns, which is more complex than the mean-variance approach used in the Markowitz model. The Markowitz model simplifies the problem by focusing on the mean and variance.

E. Ignoring the risk associated with the investments would lead to a highly risky portfolio. The Markowitz model explicitly considers risk by incorporating the variance of the returns.

6. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function in-

cludes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and discuss its practical significance in the context of ice cream production.

a. The term is not included for simplification; it adds complexity but reflects a crucial real-world cost.

b. The cost of changing production levels is a crucial component of the model because it reflects the real-world expenses associated with adjusting production lines, hiring and firing workers, and managing inventory fluctuations. In ice cream production, frequent changes are expensive and inefficient due to factors like equipment setup time, labor costs, and potential waste. Ignoring these costs would lead to an unrealistic and suboptimal production plan.

c. The term is not a placeholder; it represents a significant practical cost.

d. While related to demand, the term directly addresses production costs, not just demand fluctuations.

e. The term has a clear and important practical interpretation in the context of ice cream production.

7. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and justify why this reformulation is necessary.

   a. Using an average function introduces significant approximation error and doesn't guarantee an optimal solution.

   b. Piecewise linear approximation is possible but is not the method used in the slides; it's also computationally more complex than the auxiliary variable approach.

   c. The reformulation uses an auxiliary variable (y_i) and additional constraints to replace the max function. The constraint $a_i - x \leq y_i$ ensures that y_i is at least as large as the difference between the required hours (a_i) and the regular hours (x). The non-negativity of y_i ensures that y_i represents the number of temporary hours needed. This linearization is necessary because standard linear programming algorithms (like the simplex method) require a linear objective function and linear constraints. Non-linear functions like the max function prevent the direct application of these algorithms.

   d. Ignoring the max function leads to a significant error and an incorrect solution.

   e. Dynamic programming is not used in the slides; the problem is successfully reformulated as a linear program.

8. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

   a. Using worst-case values is overly conservative and may lead to suboptimal solutions.

   b. Ignoring the randomness completely ignores a crucial aspect of the problem, leading to an unrealistic and potentially disastrous solution.

   c. The transformation replaces the random returns (R_j) with their expected values (ER_j), converting the problem into a deterministic quadratic program. This simplification assumes that the investor is risk-neutral or that the expected return is the primary concern. The risk is then incorporated by adding a term representing the variance or covariance of the returns, leading to a quadratic objective function. This approach is widely used in portfolio optimization, but it's important to acknowledge the assumption of risk neutrality or the focus on expected return.

d. Monte Carlo simulation is a valid approach but is not the method used in the slides to transform the problem into a solvable optimization problem.

e. Using median values is a more robust approach than using expected values, but it's not the method used in the slides.

9. The Cutting Stock Problem (Slides 25-30) involves finding the optimal way to cut large rolls of paper into smaller rolls to meet customer orders while minimizing waste. Explain why this problem is typically formulated as an integer programming problem, and what challenges this introduces compared to standard linear programming.

   a. The Cutting Stock Problem is formulated as an integer program because the decision variables represent the number of times each cutting pattern is used, and these must be integers (you can't use a fraction of a cutting pattern). This integer constraint significantly increases the computational complexity compared to linear programming. Linear programming problems can be solved in polynomial time using algorithms like the simplex method or interior-point methods. Integer programming problems, however, are NP-hard, meaning that there is no known algorithm that can solve them in polynomial time in the worst case. This means that solving large-scale cutting stock problems can be computationally very expensive and time-consuming.

   b. Integer programming makes the problem harder, not easier, to solve.

   c. The discrete nature of cutting patterns is a key reason for the integer programming formulation, and it does introduce significant computational challenges.

   d. Minimizing waste is the objective, but the integer constraint is the source of the computational difficulty.

   e. Meeting customer orders exactly is a constraint, but the integer nature of the decision variables is what makes it an integer program and computationally challenging.

10. a. In the Icicle World Ltd. ice cream production planning problem (Slides 1-6), the objective function includes a term representing the cost of changing production levels between consecutive months. Explain why this term is included, and what its practical significance is in the context of ice cream production.

   A. Option A correctly identifies the reason for including the cost of changing production levels. Frequent adjustments in production are costly due to factors like equipment setup, labor scheduling, and inventory management. The model reflects these real-world costs.

   B. Option B is incorrect. The term has significant practical implications for production efficiency and cost.

   C. Option C is incorrect. The model aims to optimize production levels based on demand, not to maintain constant production.

   D. Option D is incorrect. Spoilage is a separate concern that could be modeled with additional constraints or terms, but it's not the primary reason for this term.

   E. Option E is incorrect. The term is a crucial component of the model, reflecting real-world costs.

   b. In the sales force planning problem (Slides 7-8), the original objective function contains a "max" function, making it non-linear. Explain the method used to reformulate this problem as a linear program, and why this reformulation is necessary.

   A. Option A correctly describes the linearization technique. Auxiliary variables are introduced to represent the output of the max function, effectively replacing the non-linearity with linear constraints. This is crucial because standard linear programming algorithms require linearity.

B. Option B is incorrect. Logarithmic transformations are not typically used to linearize max functions in this context.

C. Option C is incorrect. Ignoring the max function would lead to an incorrect and potentially infeasible solution.

D. Option D is incorrect. While piecewise linear approximations are possible, the method described in the slides uses auxiliary variables for a direct linearization.

E. Option E is incorrect. Standard linear programming algorithms cannot handle non-linear objective functions.

c. The Markowitz portfolio selection model (Slides 9-16) initially involves a random variable representing the return of each investment. Explain how this model is transformed into a solvable optimization problem, and what assumptions or approximations are made in this transformation.

A. Option A is partially correct in using expected values, but it incorrectly states that the result is a linear program; it's a quadratic program.

B. Option B correctly describes the transformation. The random returns are replaced by their expected values, and the variance (or a related risk measure) is incorporated into the objective function, leading to a quadratic programming problem. The assumption of independence is not strictly necessary but simplifies the model.

C. Option C is incorrect. Ignoring the randomness would lead to a highly unrealistic and potentially misleading model.

D. Option D is incorrect. While Monte Carlo simulation can be used in portfolio optimization, the slides describe a deterministic transformation using expected values and variance.

E. Option E is incorrect. Linear approximations are not typically used in this context; the variance is directly incorporated into the model.